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Review

The Transformation of Rice Crop Technology in Indonesia: Innovation and Sustainable Food Security

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Abstract: The growth of the Indonesian population has led to an increase in the demand for rice, which the country has yet to satisfy. Indonesia needs a comprehensive strategy that integrates meaningful efforts to increase its agricultural production. This study aims to review the examined trends in rice yield in Indonesia for 70 years after Indonesia's independence (1945–2016) followed by the identification of the application technology and factors that contribute to increasing rice yields to forecast sustainable food security scenarios up to 2030. This article reviews the results of research on rice production technology in Indonesia from 1945 to 2016, and the outlook for 2030. This paper examines the main points of the Indonesian transformation of rice technology: improvement of rice varieties, integrated crop management, innovations in agricultural machinery, and the Integrated Cropping Calendar Information System (ICCIS). We found that transformation has helped Indonesia increased its rice yields from 3 t ha^{-1} prior to 1961 to 4.6 t ha^{-1} in 1985, stagnated in 1990, and increased again in 2017 to 5.46 t ha^{-1}. The increase in yield was sustained by an increase in the harvested area owing to cropping index (CI) innovation. Food security and sustainable development remain the primary goals of Indonesia's agricultural sector. The application of appropriate technologies and institutional innovations can assist Indonesia in achieving its food security. Therefore, the transformation of technological innovations will continue to be an essential driver of future agricultural growth, including greater use of crop varieties, machinery, and land/institutional reforms.

Keywords: crop technology; rice; cropping index; integrated crop management; food security

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1. Introduction

Rice is the Indonesian staple food with increasing demand in line with the growth of Indonesia's population. During 2008–2020, there was a 4.66 million tons increase in rice demand or 1.16% annual average growth. The increase in rice production rate during this period was smaller, at 0.81% annual average growth. As a result of increased consumer demand, the gap between rice production and consumption has decreased by 4.32% annually.

Indonesia ranks fourth in the global population [1]. Population projections of Asia's main rice-producing and consuming countries in 1995–2025 show that Indonesia will have a population of 265 million, thus, raising an issue of how the current annual rice production can be increased from 538 million tonnes to >700 million tonnes in 2025 [2].

An increase in consumption will not be a problem if a production increase follows it. Increased rice production can be achieved by increasing the harvest index [3–6] and Cropping index (CI) through the CI Rice 400 project [7,8]. Increasing the harvest index is a solution to increase rice production in areas that do not allow the development of new agricultural land [9] and in areas where there is a lot of conversion of agricultural land to non-agriculture [10]. The conversion of agricultural land to non-agriculture in Indonesia in from 1979 to1999 was around 1,002,005 ha [11]. Indonesia needs to expand its rice area yearly to keep up with the 1.7% annual population growth [12]. Indonesia has around 9.7 million ha of potential land for expanding rice cultivation, consisting of 5.3 million ha of wetlands, 3.0 million ha of swampland, and 1.4 million ha of dry land. The policy scenario to increase the CI by around 250–300% will produce 13.25–15.9 million ha of agricultural land with a productivity of 4.2 t ha^{-1} (dry grain) and a total production of 56.31–66.78 million tons. The area of rice production in Indonesia has the potential to increase by around 11.5 million ha, 10 million ha of which are technically irrigated paddy fields [13]. Indonesia needs a comprehensive agricultural and rural development strategy centred on integrated efforts to increase agricultural production, rural infrastructure, and social and economic institutions specifically for rice.

The history of the transformation of Indonesian rice technology began when a traditional Javanese kingdom during the Dutch colonial era turned rice into a staple food. Indonesia started the transformation of the rice program by applying advanced technology to boost rice production in the 1960s through the BIMAS program [14]. The program was later upgraded to INMAS, INSUS, SUPRAINSUS, Gema Palagung, Rice Index 300, and currently, the concept of Integrated Crop Management (ICM) is applied [15]. The program to increase rice production peaked in 1984 when Indonesia was self-sufficient in rice. This article examines the trend of technological transformation and rice production in Indonesia during the 70 years after Indonesia's independence (1945–2016).

This study aims to review the examined trends in rice yield in Indonesia for 70 years after Indonesia's independence (1945–2016) followed by the identification of the application technology and factors that contribute to increasing rice yields to forecast sustainable food security in the future.

2. Rice Production in Indonesia (1945–2017)

Indonesia's rice policy has undergone many changes and experiences since its independence in 1945 [16,17]. Thus, it remains a significant issue for Indonesian people from 1945 to 1960, encouraging the government to increase food production, particularly rice [18]. Moreover, rice development from 1945 to 1960 has traditionally met daily needs [19]. Therefore, Indonesia has made a strong commitment since the 1960s to achieve rice self-sufficiency in the first decade.

Since 1961, several efforts have been made to improve the productivity and area of rice fields in Indonesia. By 1965, West Java was the largest rice producer in terms of demand and supply [20]. Rice production increased significantly in 1965 with the discovery of superior varieties with properties such as good height (tall), good breeding, early maturity, and responsiveness to fertilization [21].

From 1970 to 1985, rice yields increased significantly as a result of the early adoption of green revolution technology, combined with a stable or slight increase in area, resulting in a significant increase in national rice production [13]. However, due to constraints in the 1990–2006 production year, the national rice supply has been consistently exceeded by demand, especially in years with climatic abnormalities such as those caused by El Niño or major pest outbreaks that forced the government to import rice. Consequently, rice imports fluctuated sharply, from 0.02 million tonnes in 1993 to 4.7 million tonnes in 1999, in line with

the 'typical El-Niño-year' [13]. After 2009, Indonesia recorded the highest rice production in 2013, totalling 71.28 million tonnes, from 69.06 million tonnes in 2012. In 2010, it was 65.98 million tonnes. The increase in rice production is not solely based on considerations of superior seeds, fertilizers, appropriate growing times, and the construction of irrigation infrastructure, but also on the interaction between the planted area and productivity [22].

Nearly all rice studies have shown a significant increase in rice production in Indonesia over the past 50 years because of the successful implementation of the government's rice intensification policies and programs. Rice intensification programs should be underpinned by improved irrigation facilities and stable market conditions. Continuous institutional support was seen as a driving force behind the intensified rice revitalization. In addition, Indonesia can achieve sustainable rice production, as was the case in 1980, and alter the downward trend in farmers' incomes by (1) increasing the local government's role, (2) carefully thinking about the implementation stages, (3) producing seeds at specific locations, (4) reducing the yield gap, (5) increasing the efficiency of inputs, (6) expanding marginal land, and (7) applying information technology [13].

The following figures present data on Indonesia's rice production between 1945 and 2016 (Figure 1). R^2 was counted with the formula below (RGD Steel, JH Torri and DA Dickey. 1997. Principles and Procedures of Statistics: Biometrical Approach. Publisher: McGraw Hill):

$$R^2 = 1 - \frac{SS\ Error}{SS\ Total} = 1 - \frac{\Sigma\,(y_i - \hat{y}_i)^2}{\Sigma\,(y_i - \bar{y})^2}$$

with: y_i = observation of response i; $\hat{y}_i$ = means; and $\bar{y}$ = prediction of the i response.

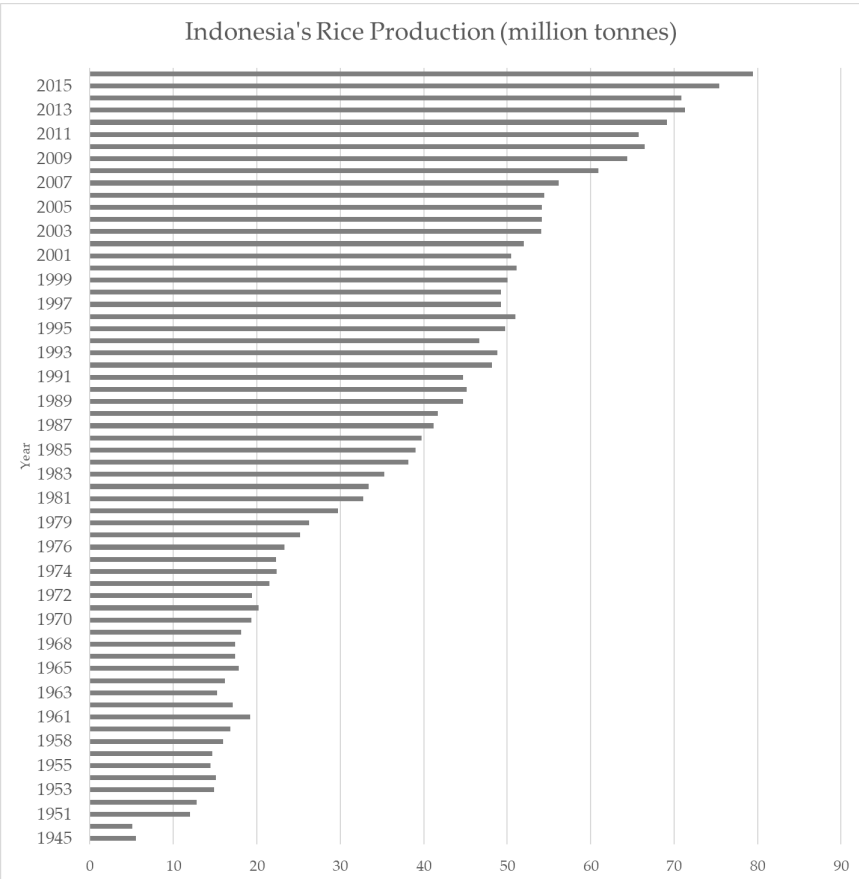


Figure 1. The Indonesian rice production from 1945 to 2016.

Rice production in Indonesia increased during the period of 1945–2017, from to 1945–1965 to R^2 = 0.7013, 1967–1983 to R^2 = 0.9119, 1984–1998 to

R^2 = 0.9053 and 1999–2016 to R^2 = 0.9426 the highest increase in rice production occurred from 2006 to 2016. Rice productivity increased from 2.8 tonnes ha^{-1} in 1976 to 5.2 tonnes ha^{-1} in 2017 with R^2 = 0.8841 (Figure 2). The rice demand scenario from 2008 to 2017 is still below the rice production in Indonesia, resulting in an annual surplus of 1.89–2.89 million tonnes (Figure 3).

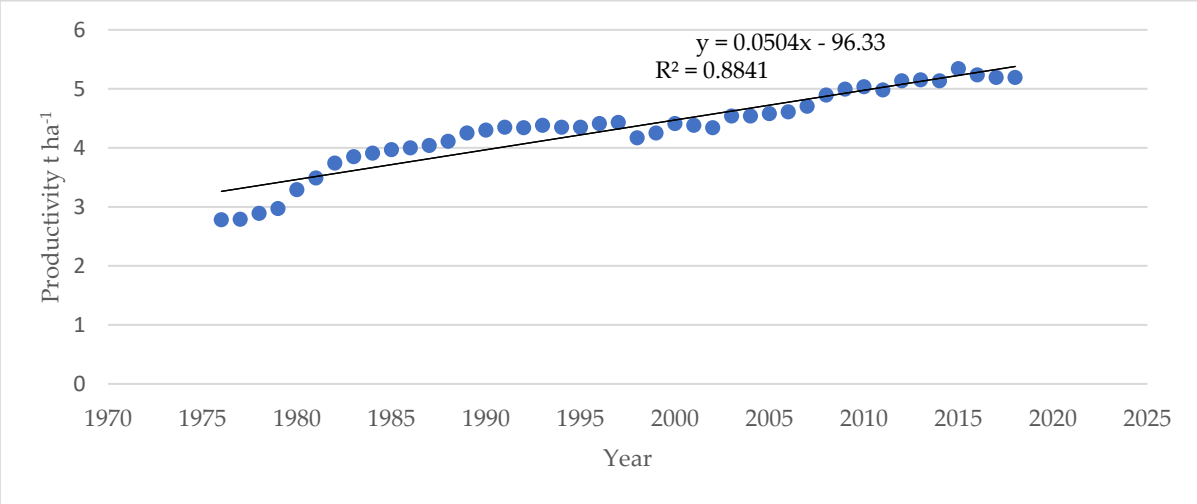
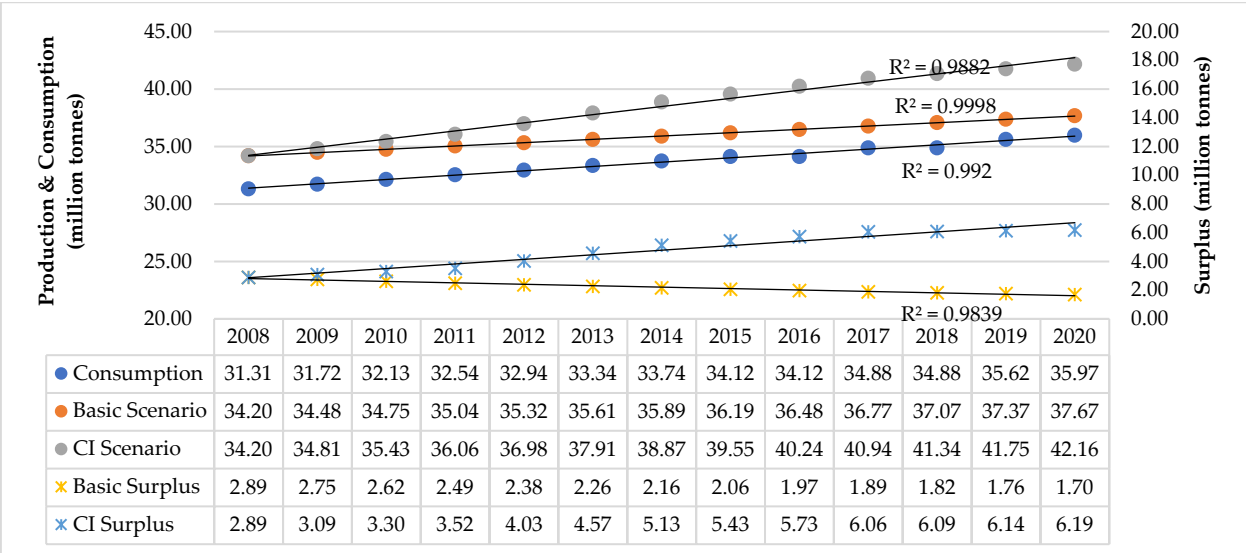


Figure 2. The average productivity of Indonesian rice in 1976–2016.



	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Consumption	31.31	31.72	32.13	32.54	32.94	33.34	33.74	34.12	34.12	34.88	34.88	35.62	35.97
Basic Scenario	34.20	34.48	34.75	35.04	35.32	35.61	35.89	36.19	36.48	36.77	37.07	37.37	37.67
CI Scenario	34.20	34.81	35.43	36.06	36.98	37.91	38.87	39.55	40.24	40.94	41.34	41.75	42.16
Basic Surplus	2.89	2.75	2.62	2.49	2.38	2.26	2.16	2.06	1.97	1.89	1.82	1.76	1.70
CI Surplus	2.89	3.09	3.30	3.52	4.03	4.57	5.13	5.43	5.73	6.06	6.09	6.14	6.19

Figure 3. The scenarios of consumption needs, CI, and surplus application of rice self-sufficiency in Indonesia (2008–2020). Note: Consumption = number of populations × per capita rice need; Cropping index (CI) scenario = production × harvested area; Basic surplus = production of CI scenario subtracted by consumption; CI surplus = production subtracted by consumption.

3. Transformation of Rice Crop Technology in Indonesia

The contribution of the agricultural sector to value addition and employment creation has generally changed with the development of agricultural innovation [13]. For example, China has made remarkable progress in feeding 22% of the world's population [23]. Consequently, the agricultural technology transformation process must be phased in to ensure that the rural economic transformation is on track for sustainable food security. Therefore, government and private institutions associated with credit, inputs, and prices directly influence adoption, use, and yield levels [2]. Support for research and extension

will be able to guarantees efficient transformation of the rice yield gap. The adoption of these improved technologies by farmers depends on the capacity of national agricultural research centres and extension services, which require additional government resources and training.

Technological innovation is instrumental to increasing rice production [24]. However, the widespread dissemination of technology for rapid adaptation by farmers has been impeded. The implementation undertaken by the government and private sector at the end of this decade has been referred to as the National Rice Production Increase Program [25]. Technological innovation used premium superior hybrid rice varieties with ICM, Field School of ICM, and direct support of superior hybrid seeds until 2019. Different technological components contribute to increased production [26]. The Food and Agriculture Organization (2000) recommends that the application of nitrogen to fertilizers is vital for rice production and productivity. Proper dosage and rapid fertilizer application contributed to high rice yield. Good quality fertilizers must be accessible to farmers. Consequently, there should be quality assurance, and its distribution should reach small-scale vendors in the village market.

3.1. Improvement of Rice Varieties in Indonesia 1945–2020

Rice crossbreeding in Indonesia began in the 1920s using a genetic pool constructed by the introduction of plants. Until the 1970s, the development program for high-yielding lowland rice varieties was more focused on improving local varieties, mainly to shorten the plant lifespan for three harvests per year [27]. The first rice variety introduced in 1943 was the Bengawan variety [28], which was an improvement over the rice varieties of China, Latisail, India, and Benong, Indonesia [28]. The Bengawan variety can be harvested 140–155 days after sowing (DAS) at 145–165 cm in height, fertilizer-responsive, and good taste, with a yield of approximately 3.50–4.0 t ha^{-1} [29]. Examples of other varieties after Bengawan include Bengawan Sigadis (1943), Jelita (1955), Dara (1960), Sinta (1963), Bathara (1965), and Dewi Ratih (1969) [27,29–31].

In 1967–1968, two introduced varieties were released, PB8 in 1967 and PB5 in 1968, in addition to the genetic source to improve the traits of the existing varieties with a potential yield of 4.50–5.50 t/ha. Crosses between PB5 and Sinta produced Pelita I-1 and Pelita I-2, whereas crosses between Pelita I-1 and Pelita I-2 produced new strains of Cisadane and Sintanur [28].

In the 1970s, high-yielding varieties played a dominant role in increasing rice production. The New Superior Varieties (NSV) contributed 56% to the increased productivity of domestic rice production [32], and the interaction between irrigation water, NSV, and fertilization contributed 75% to the increase in rice production rate [33]. In 1970–1984, the government introduced high-yielding rice varieties, such as PB5 and PB8 [34] with yields of 4.5 to 5.5 tonnes of harvested dry gain (HDG)/ha. The moisture content of HDG was 18–22%.

The variety IR64 was introduced and marketed as a superior Indonesian variety in 1986. This variety is highly appreciated by farmers and consumers primarily for its good taste, early maturity, and relatively high yield. The IR64 varieties are medium-aged (100–125 DAS), low to medium height (95–115 cm), upright shape, upright leaf position, high number of productive tillers (15–16 tillers/clump), medium panicle length, fertilizer-responsive, high-yielding (5–6 t ha^{-1}), resistant to major pests and diseases, good milled quality, and good taste [35].

In 1989, the IRRI developed and assembled rice with a new architecture, later known as the New Plant Type (NPT) [36]. Local Indonesian rice varieties of the subspecies Javanica (tropical japonica), such as Genjah Wangkal, Ketan Lumbu, and Soponyono, have been used as genetic sources or parents because of their sturdy stems, few tillers, long panicles, and large number of grains per panicle [33,36].

The development of NSV in Indonesia began in 1995 [37] with four varieties were later introduced, namely Cimelati (2001), Gilirang (2002), Ciapus (2003), and Fatmawati

(2003). However, all four varieties have drawbacks, including high empty grain content and low resistance to major pests and diseases. This requires the assembly of a New Type of Superior Variety with a higher yield potential, increased resistance to major pests and diseases, and improved quality [38].

In 2001, NSV research became more intense with the introduction of more NPT: IRRI 1 and second-generation elite lines, IRRI 2 (the crossing of NPT IRRI 1 and Indica rice lines). The selection and crossing of new varieties used IRRI 1 and IRRI 2 [37]. The developed varieties have an improved rice taste (more delicious, fluffier) and are resistant to plant-disturbing organisms. Over the last few decades, the purpose of assembling high-yielding varieties has been to produce high-yielding rice that can withstand environmental stresses, pests, and diseases. The varieties include Inpari and Inpara, produced by the Indonesian Center for Rice Research (ICRR) [39]. Resistant varieties can play a vital role in controlling pest attacks on rice plants [40–42].

The IR64 national planted area was 45.5% in 2002, indicating a low adoption of NSV by farmers. This requires continued efforts to disseminate NSV [43]. Several factors account for the slow development of NSV: (1) the superiority of NSV over the existing varieties, (2) lack of interest by the seed industry to develop NSV, and (3) limited supply of seed sources for commercial seed propagation [44]. NSV properties of interest to farmers include age, plant height, productive tillers, resistance to pests, and yield [45]. Productivity was no longer an adoption issue because most NSV yields were higher than those of the farmers. This showed that farmers preferred Inpari 12 less, although it showed high productivity and early maturity due to its taste [46]. The main concern was the limited availability of NSV seeds owing to a shortage of seed sources [47].

In 2012, the Ministry of Agriculture released 493 superior varieties, followed by 57 high-yielding rice varieties from 2010 to 2015, including 31 varieties of lowland rice, 6 varieties of swamp rice, 9 varieties of highland rice, and 11 varieties of hybrid rice [48]. The latest data from 2014 showed that Ciherang still dominated rice cultivation in South Sumatra, with an area of 635,195 ha (74.83% of the planted area), Ciliwung variety 72,867 ha, local varieties 68,780 ha, IR42 29,187 ha, and other varieties less than 14,000 ha. Even until 2013, Ciherang was the dominant rice cultivated in Indonesia, with a 37.1% planted area [49]. Subsequently, 35 NSVs were released between 2016 and 2020 [50].

3.2. Integrated Crop Management (ICM)

A limited-scale concept test was conducted in Indonesia, between 1996 and 1997. The holistic integration of crop management practices is essential in Integrated Crop Management Packages (ICMPs) with a flexible approach to adapting to existing environmental, socioeconomic, and market factors [2]. Integrated Crop Management (ICM) practices encompass approaches to managing land, climate, water, crops, pests, and plant diseases to increase productivity, farm income, and environmental sustainability. The four principles of ICM are integration, interaction, dynamics, and participation. The components of ICM are basic technology and the selection of technology components. The basic technology consists of (1) modern varieties, (2) healthy and high-quality seeds, (3) efficient fertilization, (4) ICM according to target pest, and the selection of technological components consisting of (1) crop management, (2) early age paddy seedlings, (3) organic fertilizers, (4) intermittent irrigation, (5) liquid fertilizers, and (6) harvest and post-harvest handling [51]. In addition, political policies, especially prices, have a strong influence on production [52].

The results of ICM application in irrigated rice fields at the experimental scale (1–1.25 ha) conducted by the ICRR since 1999 indicated an average yield increase of 37%. The rate decreased to 27% at a higher assessment level (1–5 ha) and to 16% at 30 ICM locations with an area of 50–100 ha. Moreover, increased ICM grain yields and improved rice quality have reduced rice cultivation costs as well as maintaining health and environmental sustainability [53]. ICM increased production by 5–12% relative to traditional farming models. Labour shortages in future agriculture have been replaced by the introduction and

application of machine tools, from the manufacturing of nurseries, soil treatment, planting, weeding, and harvesting machinery in rice-growing centres in Indonesia.

An effort by The Indonesian Agency for Agricultural Research and Development (IAARD) to boost the production and productivity of food crops was to modify the planting system using the Jajar Legowo (Jarwo) technique in 2008. This technique adjusts the spacing to compact paddy clumps in rows, widens the space between compact rows, and optimizes growth space and sunlight. For example, in Jarwo 2:1 (2 rows-1 space), all rice clumps were planted with spaces (empty row) after every two rows to ensure that they benefitted from the boundary effect [54]. The planting system using the Jarwo technique offers advantages such as: (1) higher plant population, (2) lower pests (rats and snails) attack, and (3) reduced iron toxicity [55]

The IAARD introduced Jarwo Super Technology (JST) in 2018, which upgrades the former Jarwo technology by adding integrated crop management using biodecomposer and agricultural machinery in irrigated paddy field. As a result, JST was able to increase production to 1.7–2.97 tonnes/ha [56,57]. The adoption of Jarwo Super increased the rice yield by 12–37%, while the allocation of agricultural costs for Jarwo Super practice was 37.19% and for traditional planting farmers was 43.59%. The values for each revenue/cost (R/C) ratio were 2.69 and 2.29, while the B/C ratios were 1.69 and 1.29, with an MBCR value of 5.25, meaning that the introduction of JST was financially feasible [57,58]. However, JST requires certain fundamental aspects that must be implemented on site: (1) NSV with high yield potential, (2) bio-decomposer, given before soil processing, (3) biofertilizer and balanced fertilization based on soil testing equipment, (4) pest management using biopesticides and chemical pesticides for the control threshold, and (5) agricultural tools and machinery, especially for planting (transplanter) and harvesting (combined harvester) (IAARD, 2016). JST 2:1 application can increase yield by 1483 kg/ha, whereas JST 4:1 (4 rows-1 space) has an increased yield of 852 kg/ha compared to the non-Jarwo planting system [59]. ICM approaches [2] have the potential to accelerate yield gap bridging and increase production. Approaches include locally available technologies, active institutional support from governments (especially for inputs and village credit), and stronger research and extension linkages. Thus, the technological packages specified for the location moving towards "prescription farming" can be made available and popularized. However, the gaps in yield performance should first be understood.

3.3. Implication of Agricultural Machine Tools Innovation

The introduction of more efficient technologies for handling, drying, storing, and milling rice in villages is crucial for reducing post-production losses [2]. Innovation is a shift in the understanding, practice, methods, and habits that were previously transformed into new things, reflected in the final outcome, adoption, discovery, process, and behavioural change in the use of technology. The proper use of agricultural tools and machinery is a measurement tool that increases the production and efficiency of labour and time to become a nation's competitiveness point [60]. The requirement and availability of agricultural machine tools in Indonesia can be seen in Table 1.

Table 1. The requirement and availability of agricultural machine tools in Indonesia (2010–2016).

Year	Tractor (2 Wheels)		Tractor (4 Wheels)		Water Pump		Rice Transplanter	
	Requirement	Availability	Requirement	Availability	Requirement	Availability	Requirement	Availability
2010	320,012	4.036	177,834	7	533,503	3622	400,128	0
2011	323,838	662	17,991	11	539,731	410	404,798	176
2012	325,091	1.567	180,606	50	541,818	600	406,363	0
2013	324,484	3.996	180,269	0	540,807	2002	405,605	153
2014	325,977	15.435	181,098	0	543,295	4069	407,471	379
2015	327,476	25.509	181,931	1244	545,794	16,271	409,345	507
2016	328,982	262.090	182,768	2822	548,304	354,699	411,228	8601

Using agricultural machinery can reduce farming costs and provide farmers with benefits that can help them achieve food self-sufficiency. Furthermore, agricultural mechanization has good prospects if it is preceded by a mapping of needs, available resources, and an adequate institutional environment. Therefore, it is possible to reduce agricultural costs and increase efficiency [61]. Therefore, technological innovation using agricultural machinery is crucial for advancing Indonesian agriculture [62].

Since Indonesia's independence (1945), farm equipment has undergone significant changes under human power, motor, wind, water, and other energy sources. The weakness of traditional harvesters is a yield loss of approximately 9.52–10%, with large areas of rice fields and large quantities of labour for harvesting. Over time, harvesters have become increasingly scarce because the agricultural workforce has shifted to the industrial sector, resulting in higher harvester costs. One study found that the use of conventional harvesting tools caused approximately 10% (9.52%) yield failure. The adoption of tractors increased from 15.2% in 1990 to 19.4% in 1993, whereas the adoption of rice threshers increased from 15.4% to 25.6% [63]. The gradual development of harvesting tools before soil processing tools indicated that the stages of rice cutting, collection of rice pieces, and threshing were critical points of yield loss [64].

A large area of rice field requires a large number of harvesters. Farmers may use a reaper, stripper, combine harvester, or rice mower to overcome labour shortages and reduce yield losses. A study [65] revealed that a rice mower reduced crop losses by 1.44% and was economically feasible. The break-even point value when using a rice mower machine was 1.88 ha year^{-1}. The Net Present Value (NPV) of the rice mower machine was IDR 10,232,314.18 year^{-1}, and the benefit/cost (B/C) ratio of the rice mower machine was 1192. Hand tractors, mechanical threshers, pedal threshers, and pest sprayers are all components that can be created and produced locally. This would create income opportunities for the local community and farmers, whereas owners or contractors of equipment, operators, and workshops would see an increase in revenue.

3.4. Implication of Integrated Cultivation Calendar (ICCIS), Indonesia Modern Farmer Information System

In order to increase and accelerate access to information for users, particularly extension workers and farmers, the Indonesian Ministry of Agriculture, through the IAARD, has developed a web-based information technology system and applications using smartphones, known as the ICCIS. This system is based on climatic variability and water availability for crops. ICCIS provides a potential planting time model for food crops, particularly rice, maize, and soybean [66]. Information on planting calendar estimations is supplemented with additional information, such as areas prone to flooding, droughts, pests, and disease attacks. It also provides new varieties and recommendations for the proper dosage of fertilizer, agricultural machinery, and adequacy of livestock nutrition as a reference point for decision makers in food crop policy formulation at the sub-district, regency, and provincial levels.

The future of food production is expected to become more complicated and demanding. Food production is affected by factors, such as climate variability and climate change. Indonesia Agricultural Research and Development has developed a Crop Calendar Information System (ICCIS) to adapt to climate change. The recommendations included in the ICCIS are planting time, cropping patterns, and adaptive technology integrated web-based based on climate forecasts and the season to date [67]. This technology can provide a wide range of recommendations regarding the appropriate technology to adapt to climate change and the availability of supporting innovation and current farming patterns [67,68]. The ICCIS is updated bi-annually based on two seasons in Indonesia: the wet season from October to March and the dry season from April to September. The coverage area of ICCIS is 7,459,891 ha, distributed across five major islands, those being Sumatra (1,752,308 ha), Java (3,472,864 ha), Kalimantan (723,947 ha), Sulawesi (972,854 ha), and East Indonesia (537,919 ha).

The new ICCIS feature provides a map of rice-growing stages from Sentinel-2 (S-2) images with a resolution of 10 m × 10 m. Sentinel-2 from the Copernicus Hub enables users to download the latest images of the Earth's surface every five days with two constellations of satellites at no charge. S-2 provides a multispectral imaging system that correlates the growth stages of rice. Classification of rice growth stage based on the work applied machine learning to classify bare land, vegetative, reproductive, and maturation stages of 10 bands of S-2 with high precision [69].

Machine learning technique is the basis for an automated mapping system for rice growth stages across the official Indonesian rice area of 7.1 million ha from regency into village levels, divided into 5525 sub-districts. The system also calculates the daily aggregation of village areas on a national basis. On average, the system can update information from 30 regencies on a daily basis and upload the information into cloud storage for user-friendly access. For the entire 2020 year, 12,280 regencies' maps and 81,874 sub-district maps were produced. The highest and lowest data acquisitions occurred in August 2020 and January 2020, with 15,809 and 6340 maps at the sub-district level, respectively. The application of the yield estimation model using the NDVI value was developed from linear regression analysis to estimate rice yield with a mean precision value of 80.3% [70].

4. The Role of Government Policy and Extension in The Transformation of Rice Technology

The transformation of rice technology has made a real contribution to increasing rice production and productivity from year to year. In every government policy in the program to increase rice production through new technology packages which include production and productivity lever components. For example, increasing the CI is a strategic policy as compensation for land conversion and increasing the quality of intensification through the use of new superior varieties accompanied by integrated management of plants and resources [71]. Policy implementation must be supported by the provision of various supporting facilities that enable farmers to adopt new technologies. Increasing national rice production in the context of food self-sufficiency is carried out by applying superior rice cultivation technology at the farm level. Efforts to increase rice production at the farm level are not only related to technical and economic aspects, but also a strategy to mobilize farmer participation in policy programs to realize active participation of farmers in increasing rice production in a sustainable manner and increasing farmer income. The achievement of a sustainable level of rice production is the result of the integrated participation of farmers in the application of cultivation technology and cooperation within groups supported by smooth services and counselling [72].

Extension is a means of implementing effective policies due to limited knowledge and insights on increasing rice production at the farmer level. Agricultural extension aims to increase knowledge that helps in increasing technical and production efficiency [73,74]. Starting in 1945, the number of extension agents in Indonesia has always increased until now it has reached more than 500,000 agricultural extension workers who play an important role

in increasing the competence of farmers. The role of extension workers as communicators, facilitators, educators and communicators has a positive effect on increasing the competence of farmers. Agricultural extension services are a tool that connects farmers with sources of information and seeks solutions to overcome problems in applying technology to increase production [75].

An effective extension strategy for increasing farmer participation in programs to increase rice production is participatory extension by trying to improve farmers' abilities through intensive interactions with extension agents, providing relevant agricultural information sources, conducting meaningful learning and learning from their own experiences [76,77]. Latif et al. [78] states that there is a significant relationship between the role of extension workers and farmers' perceptions of the performance of extension agents with increased rice productivity. The intensity of interaction between extension agents and farmers has a direct effect on the level of farmers' perceptions of the role of extension agents in increasing rice production [79]. Rice productivity in the working areas with extension assistance is higher than in areas without extension assistance [80,81].

Along with advances in rice technology, in addition to increasing the number of extension agents, capacity building of extension agents was also carried out, improvement of extension strategies and provision of adequate logistics to improve extension performance [82]. The extension strategy approach has also transformed which is not only oriented towards program implementation but adaptive and sustainable assistance. On the other hand, the government has also formulated comprehensive extension policies according to the needs of extension workers and farmers [83]. A sustainable increase in rice production will be achieved with synergistic integration between the government and policies and programs, the role of extension workers with counselling and mentoring strategies, as well as the active participation of farmers in the adoption and application of technology.

Agricultural Extension materials in support of increasing Food Availability, Food Access, and Food Consumption Quality are prepared by taking into account: (a) natural resource potential; (b) availability of local Food genetic resources; (c) market development potential; (d) availability of human resources; (e) availability of agricultural facilities and infrastructure; (f) planting season and harvest schedule; (g) market demand; (h) prices at the level of producers and consumers; (i) food insecurity conditions and cases of malnutrition; (j) regulations related to standardization and quality of food products; (k) local food availability and vulnerability; and (l) public interest in consuming diverse, nutritious, balanced, quality and safe food (Presidential Decree No. 35 of 2022) and strengthened the establishment of extension institutions (Regulation of Ministry of Agriculture No. 3 of 2018).

5. Conclusions

Food security and sustainable development will continue to be central to the primary goals of the Indonesian agricultural sector. One of the main challenges facing Indonesia is the way Indonesian agriculture can meet its food demand with superior quality and more varieties in a sustainable approach by 2030. Over the course of 75 years, the increase in rice production has not slowed the population growth. This means that concerns and the ability to import on an annual basis will continue to be required for reserves and price stability to avoid triggering inflation. Indonesian rice productivity remains low, on average, especially in regions outside Bali and Java. The adoption of new technologies will be effective if the production process using existing technologies is efficient, given the magnitude of land ownership. Rice cultivation on irrigated land is technically more efficient because rice grows better in wetlands than in drylands, saline, and swamps.

The context of food security in the past was simply that the agricultural sector was capable of meeting the nation's food needs. Applying appropriate technologies and institutional innovations can help ensure food security in Indonesia. The agricultural sector has performed exceptionally well in achieving this goal. The transformation of technological

innovations will continue to be an essential driver of future agricultural growth, including the increased use of crop varieties, machinery, and land/institutional reforms.

Strategic priorities were set for each zone through a clear and well-thought-out zoning process. This will reduce political conflict, secure food production, and safeguard ecologically vulnerable areas. Furthermore, it will increase the annual IC and maintain the perennial land area for rice production from 10.4 to 14,410 ha^{-1}. Parallel to the changing socioeconomic situation in Indonesia, the concept of food security has expanded in several directions. Food security includes issues related to the quantity and quality of food. For example, food security is a major challenge today and in the years to come as the country's state of soil and water resources continues to deteriorate.

In addition, as income levels increase, improved food quality and more diverse eating habits impose additional constraints and demands on the country's agricultural industry. It is appropriate to promote the conservation of agricultural species and genetics, as well as more environmentally friendly farming systems such as integrated rice–fishery, ICM, and the organic rice system. Not only do they provide a high-quality diet, but they are also good for the environment and agrobiodiversity. Enhanced technologies, such as profitable production of bioenergy, can help recycle tailings to address the issue of air and water pollution caused by crop straw or animal manure and offer new possibilities for the use of sustainable energy in rural areas.

Given the country's history of famine, food shortages, and hunger between 1945 and 1965, Indonesia has accomplished an outstanding job in meeting its food requirements. However, progress has been made with price tags. Maintaining the link between agriculture and the environment on a more sustainable path is a major challenge today and in the future. Going forward, both challenges and opportunities exist. Transformative technological innovation is the driving force behind agricultural development.

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